

Script-Controlled ACL: Restrict Record Access Based on Field Value

Date	29 September 2026
Team ID	SWTID-2026-1374
Project Name	Script-Controlled ACL – Restrict Record Access Based on Field Value

Project Description:

The main objective of this project is to implement role-based and field-based record security using ServiceNow Access Control Lists (ACLs). The solution protects a custom table, Institution Details (u_institution_details), so that every record operation – Read, Create, Write and Delete – is allowed only for users who hold the matching role.

A script-controlled READ ACL evaluates the logged-in user: admin users are allowed to see all records, users holding the bb1 role are permitted to access only the records whose Branch is EEE, and every other user is denied. Separate ACLs for Create (bb2), Write (bb3) and Delete (bb4) complete the security model, giving administrators precise control over who can view and change student data.

Tools and Technologies Used:

- ServiceNow
- ServiceNow Administration
- Access Control Lists (ACL)
- JavaScript
- ServiceNow Roles
- Custom Tables
- User Administration

Scenarios

Scenario 1: The EEE User, who holds the bb1 role, opens the Institution Details list. The script-controlled READ ACL allows access and the data condition "Branch is EEE" ensures that only the EEE records are shown, while ECE and CSE records stay hidden.

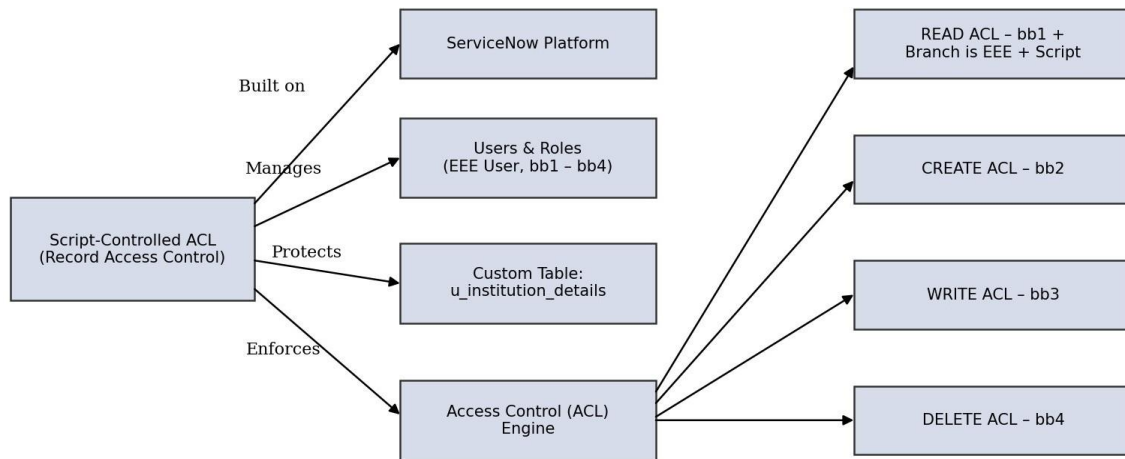
Scenario 2: An administrator opens the same table. Because the ACL script returns true for the admin role, all records – EEE, ECE and CSE – are available without any restriction.

Scenario 3: A user who has neither the admin role nor the bb1 role tries to view Institution Details. The ACL script returns false and access is denied. In the same way, only users with bb2, bb3 or bb4 can create, modify or delete records respectively.

Key Concepts:

Concept	Description
Access Control List (ACL)	Security rules used in ServiceNow to control who can access records and what operations they can perform.
READ ACL	Controls whether a user can view records.
CREATE ACL	Controls whether a user can create records.
WRITE ACL	Controls whether a user can modify records.
DELETE ACL	Controls whether a user can delete records.
Script-Controlled ACL	A script can be used to evaluate conditions and return true or false to determine whether access should be allowed.

Technical Architecture:

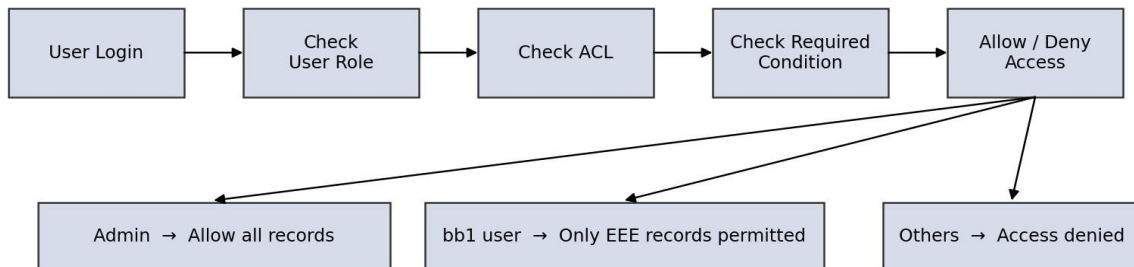


Description: The solution is built entirely on the ServiceNow platform. Users and roles (bb1 – bb4) are created through User Administration, the custom table u_institution_details stores the student records, and the Access Control (ACL) engine enforces four record-level ACLs on that table. The READ ACL is an advanced ACL that combines a required role, a data condition (Branch is EEE) and a server-side JavaScript script.

Database: The project uses a single custom table, Institution Details (u_institution_details, extends table: False), which holds the required student fields and a Branch field with the values EEE, ECE and CSE. As there is only one table, no relationships exist and a separate ER diagram is not required.

Security Logic:

The project follows this basic logic every time a user tries to work with a record:



For the READ ACL: Admin → Allow all records; bb1 user → access permitted to EEE records; Other users → Access denied.

Pre-requisites:

- **ServiceNow Administration Knowledge:** [ServiceNow Developer Program](#)
- **ACL and Security Concepts:** [ServiceNow Documentation](#)
- **JavaScript Basics:** Understanding of functions, conditions and return values for the ACL script.
- **User and Role Management:** Creating users, creating roles and assigning roles.
- **Custom Tables:** Creating tables, fields and records in ServiceNow.
- **Access to a ServiceNow Instance:** An instance with the admin role to configure users, roles, tables and ACLs.

Project Workflow:

Milestone 1: User and Role Creation

- **Activity 1.1:** Create the test user EEE User in User Administration.
- **Activity 1.2:** Create the roles bb1, bb2, bb3 and bb4.
- **Activity 1.3:** Assign the required roles to the EEE User.

Milestone 2: Table Creation and Data Preparation

- **Activity 2.1:** Create the custom table Institution Details (u_institution_details).
- **Activity 2.2:** Add the required student fields to the table.
- **Activity 2.3:** Create multiple records with different Branch values such as EEE, ECE and CSE.

Milestone 3: ACL Configuration

- **Activity 3.1:** Create the script-controlled READ ACL (role bb1, data condition Branch is EEE).
- **Activity 3.2:** Create the CREATE ACL (role bb2).
- **Activity 3.3:** Create the WRITE ACL (role bb3).
- **Activity 3.4:** Create the DELETE ACL (role bb4).

Milestone 4: Testing and Verification

- **Activity 4.1:** Test the READ ACL as EEE User, as an admin user and as a user without the bb1 role.
- **Activity 4.2:** Verify that the Create, Write and Delete ACLs allow only bb2, bb3 and bb4 users respectively.

Milestone 5: Conclusion

Milestone 1: User and Role Creation

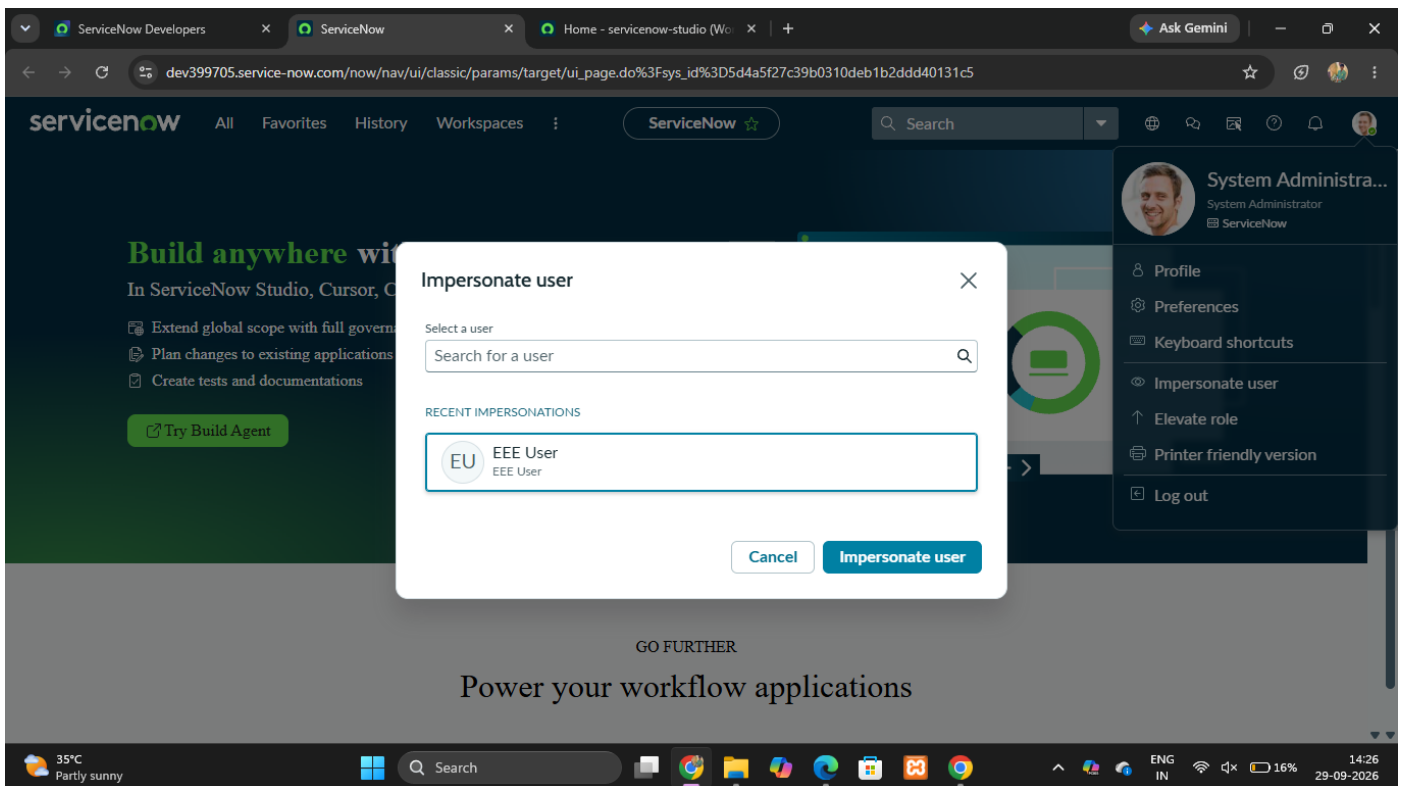
In this milestone, the test user and the four roles that drive the whole security model are created. Each role is later linked to exactly one record operation through its ACL.

Activity 1.1: Create the Test User

Navigate to User Administration → Users → New and create the following user:

Field	Value
User ID	EEE User
First Name	EEE
Last Name	User
Email	eeeuser@gmail.com

Click Submit / Save to create the user.



Activity 1.2: Create the Roles

Navigate to User Administration → Roles → New and create the four roles listed below:

Role	Purpose in this project
bb1	Required for the script-controlled READ ACL (view EEE records)
bb2	Required for the CREATE ACL (create records)
bb3	Required for the WRITE ACL (modify records)
bb4	Required for the DELETE ACL (delete records)

The screenshot shows the ServiceNow 'Edit Members' page for the 'User Role'. The browser address bar displays the URL: `dev399705.service-now.com/now/nav/ui/classic/params/target/sys_m2m_template.do%3Fsys_is_list%3Dtrue%26sys_is_related_list%3Dtrue%26sys_target%3Dsys_user_...`. The page header includes the ServiceNow logo and navigation tabs: 'All', 'Favorites', 'History', 'Admin', and 'User Role - Edit Members'. A search bar is also present.

The main content area features a filter section with 'Add Filter' and 'Run filter' buttons. Below this, there are two lists: 'Collection' and 'Roles List'. The 'Collection' list contains the following items: 'bb1', 'bb2', 'bb3', and 'bb4'. The 'Roles List' is currently empty, showing only '--None--'. Navigation arrows (left and right) are located between the two lists. At the bottom of the filter section, there are 'Cancel' and 'Save' buttons.

The Windows taskbar at the bottom shows the system clock as 14:28 on 29-09-2026, with a battery level of 16% and network status 'ENG IN'.

Activity 1.3: Assign Roles to the EEE User

Open the EEE User record and add the required roles under the Roles related list, then save the record.

This screenshot shows the same ServiceNow 'Edit Members' page, but with the roles assigned to the 'EEE User'. The 'Collection' list remains empty, showing only '--None--'. The 'Roles List' now contains the following items: 'bb1', 'bb2', 'bb3', and 'bb4', which are highlighted in blue. The 'Cancel' and 'Save' buttons are still visible at the bottom of the filter section.

The Windows taskbar at the bottom shows the system clock as 14:28 on 29-09-2026, with a battery level of 16% and network status 'ENG IN'.

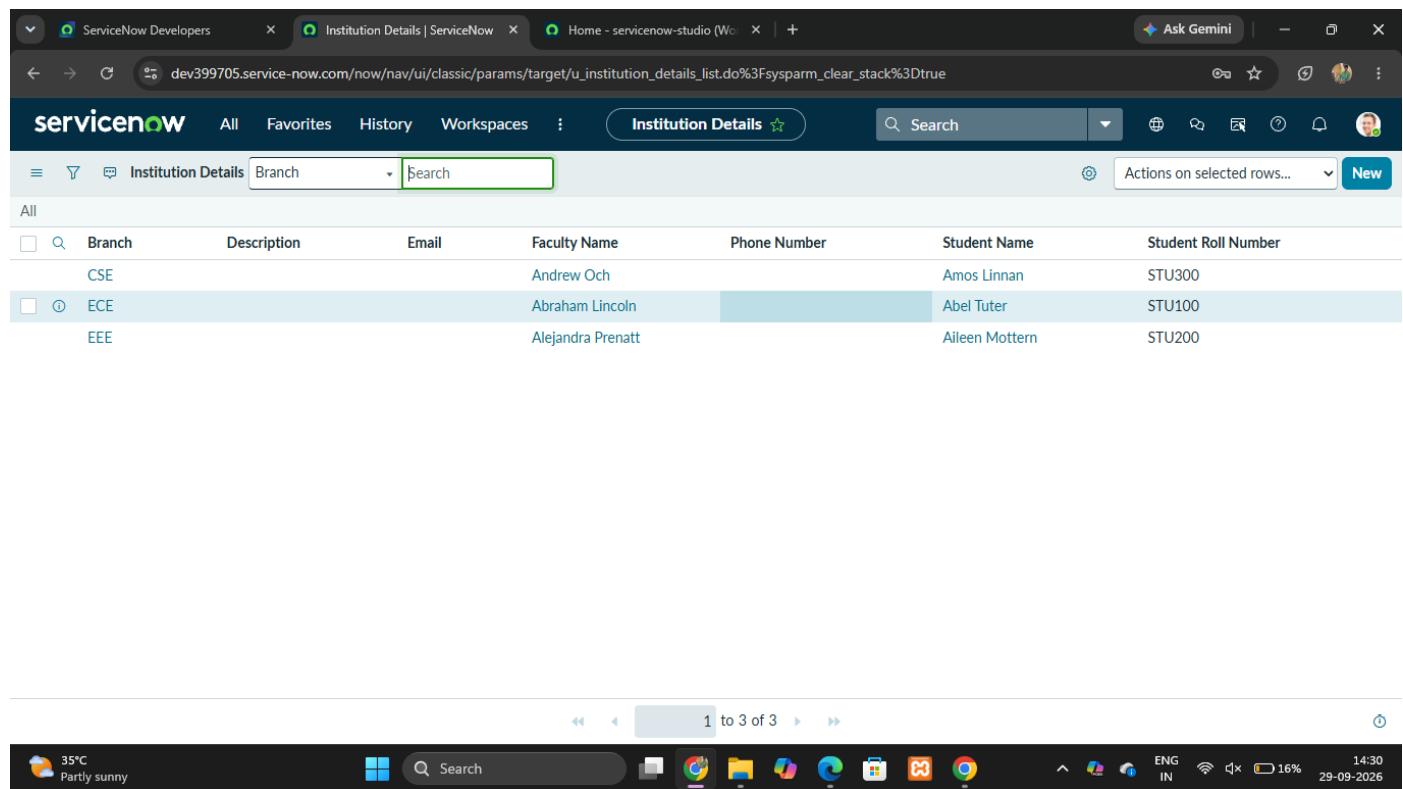
Milestone 2: Table Creation and Data Preparation

This milestone creates the custom table that will be protected by the ACLs and fills it with sample records so that field-based restriction can be verified.

Activity 2.1: Create the Custom Table

Navigate to System Definition → Tables → New and configure:

Field	Value
Label	Institution Details
Name	u_institution_details
Extends table	False



The screenshot shows the ServiceNow interface for the 'Institution Details' table. The table has columns: Branch, Description, Email, Faculty Name, Phone Number, Student Name, and Student Roll Number. The data is as follows:

Branch	Description	Email	Faculty Name	Phone Number	Student Name	Student Roll Number
CSE			Andrew Och		Amos Linnan	STU300
ECE			Abraham Lincoln		Abel Tuter	STU100
EEE			Alejandra Prenatt		Aileen Mottern	STU200

The interface also shows a search bar, a 'New' button, and a status bar at the bottom indicating 1 to 3 of 3 records.

Activity 2.2: Add the Student Fields

Add the required student fields to the table, including the Branch field that the READ ACL uses for its data condition.

Activity 2.3: Create Sample Records

Create multiple records with different Branch values such as EEE, ECE and CSE, so that the effect of the "Branch is EEE" condition can be tested.

ServiceNow DevelopersInstitution Details | ServiceNowHome - servicenow-studio (Win)Ask Gemini

dev399705.service-now.com/now/nav/ui/classic/params/target/u_institution_details_list.do%3Fsysparm_clear_stack%3Dtrue

servicenowAllFavoritesHistoryWorkspacesInstitution Details ☆Search

Institution Details BranchSearchActions on selected rows...New

All

☐	Branch	Description	Email	Faculty Name	Phone Number	Student Name	Student Roll Number
☐	CSE			Andrew Och		Amos Linnan	STU300
☐	ECE			Abraham Lincoln		Abel Tuter	STU100
☐	EEE			Alejandra Prenatt		Aileen Mottern	STU200

1 to 3 of 3

35°C
Partly sunny

Search

ENG
IN

16%

14:30
29-09-2026

Milestone 3: ACL Configuration

In this milestone, four record-level ACLs are created on the u_institution_details table. Navigate to System Security → Access Control (ACL) → New for each ACL.

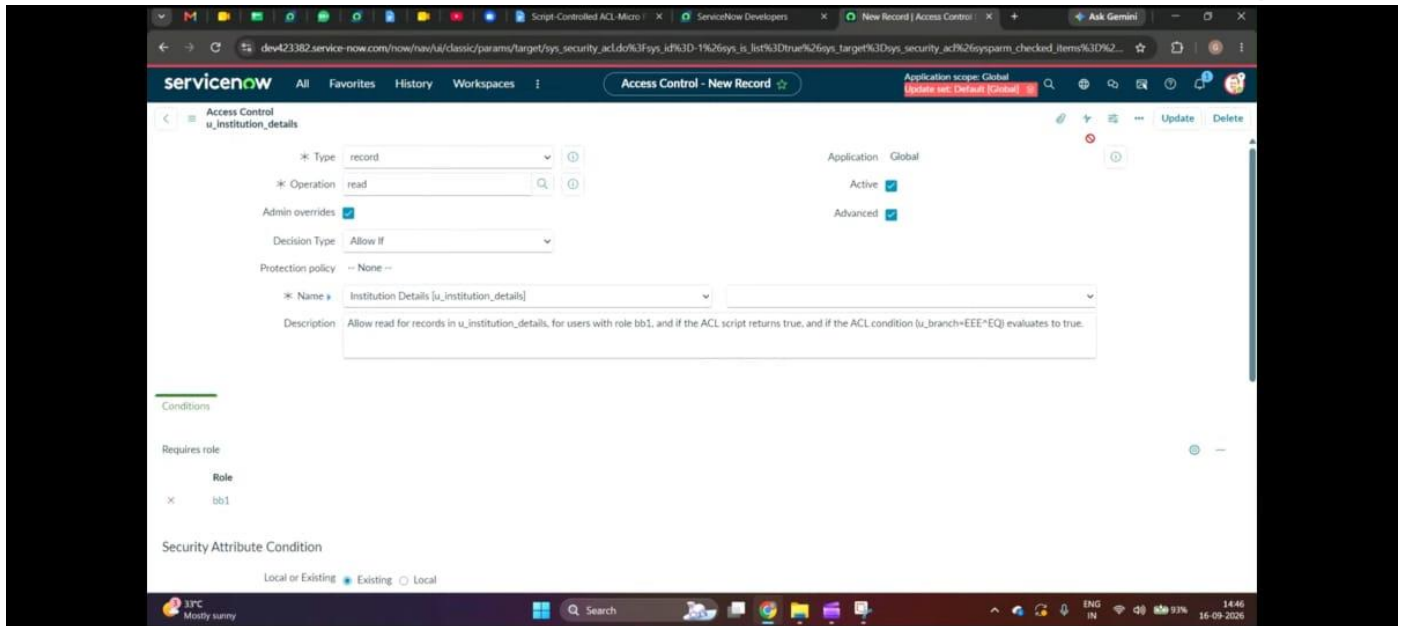
Activity 3.1: Create the READ ACL

Field	Value
Type	Record
Operation	Read
Name	u_institution_details
Active	True
Advanced	True
Required role	bb1
Data condition	Branch is EEE

Add the following script in the Script field and save the ACL:

```
(function () {  
  // Allow admin users full access  
  if (gs.hasRole('admin')) {  
    return true;  
  }  
  
  // Allow users with bb1 role  
  if (gs.hasRole('bb1')) {  
    return true;  
  }  
  
  // Deny access for all others  
  return false;  
}) ();
```

- Admin users are allowed full access.
- Users with the bb1 role are allowed, and the data condition limits them to EEE records.
- All other users are denied, because the script returns false.



The screenshot displays the ServiceNow 'Access Control - New Record' configuration page. The browser address bar shows a URL from dev423382.service-now.com. The page title is 'Access Control - New Record'. The configuration is for an 'Access Control' record named 'u_institution_details'.

Configuration Details:

- Type:** record
- Operation:** read
- Application:** Global
- Active:** ☒
- Advanced:** ☒
- Admin overrides:** ☒
- Decision Type:** Allow If
- Protection policy:** -- None --
- Name:** Institution Details [u_institution_details]
- Description:** Allow read for records in u_institution_details, for users with role bb1, and if the ACL script returns true, and if the ACL condition (u_branch=EEE*EQ) evaluates to true.

Conditions:

- Requires role:**
- Role:** bb1

Security Attribute Condition:

Local or Existing ☒ Existing ☐ Local

Activity 3.2: Create the CREATE ACL

Field	Value
Type	Record
Operation	Create
Name	u_institution_details
Active	True
Required role	bb2
Data condition	Not required

Activity 3.3: Create the WRITE ACL

Field	Value
Type	Record
Operation	Write
Name	u_institution_details
Active	True
Required role	bb3
Data condition	Not required

Activity 3.4: Create the DELETE ACL

Field	Value
Type	Record
Operation	Delete
Name	u_institution_details
Active	True
Required role	bb4
Data condition	Not required

ACL Summary

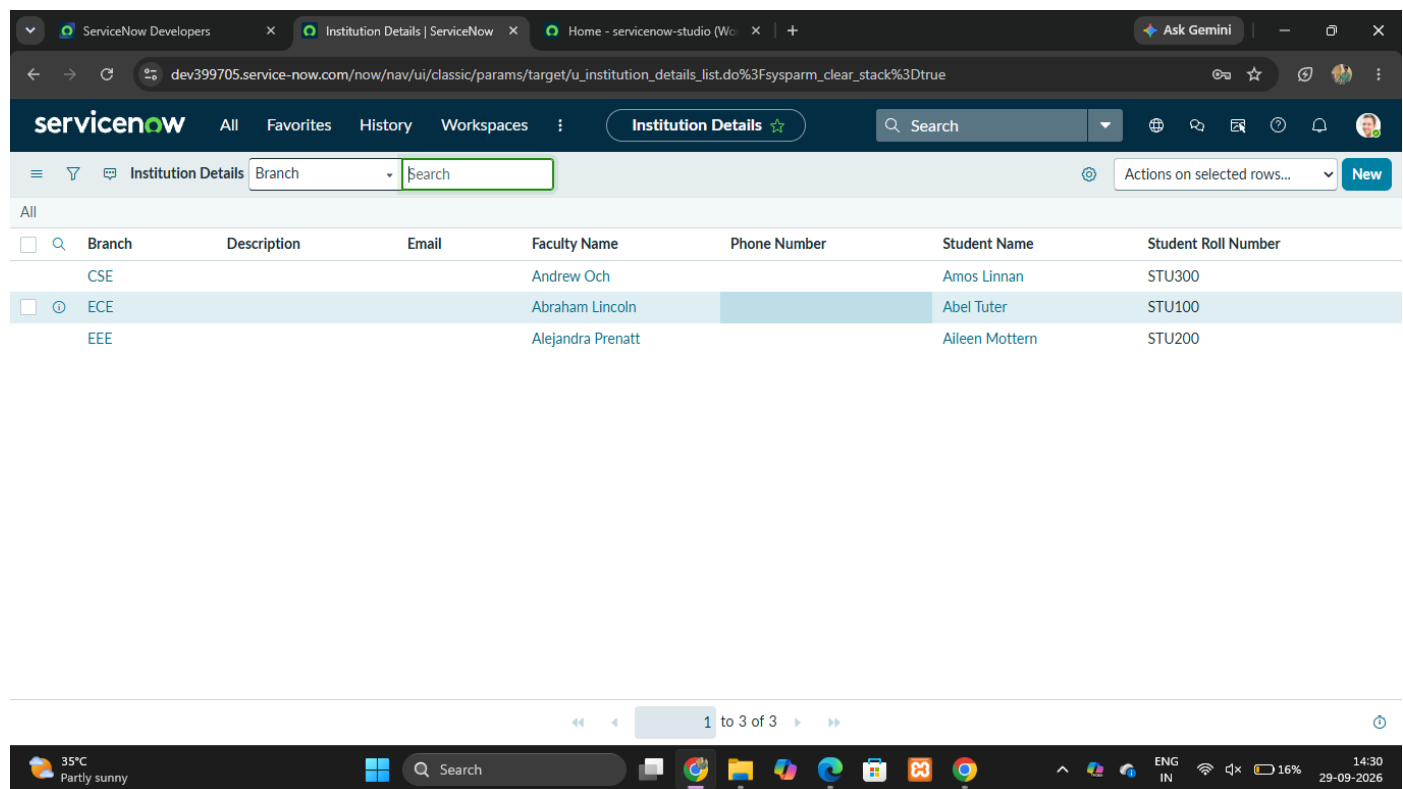
ACL	Operation / Required Role	Advanced	Data Condition / Script
READ ACL	Read / bb1	True	Branch is EEE + script (admin and bb1 allowed)
CREATE ACL	Create / bb2	No	None
WRITE ACL	Write / bb3	No	None
DELETE ACL	Delete / bb4	No	None

Milestone 4: Testing and Verification

After configuring the ACLs, the behaviour is verified by impersonating the different users and opening the Institution Details table.

Activity 4.1: Verify the READ ACL

User	Expected Result
Admin user	All records (EEE, ECE and CSE) are visible
EEE User (bb1)	Only records with Branch = EEE are visible
User without bb1	Access denied – no records are visible



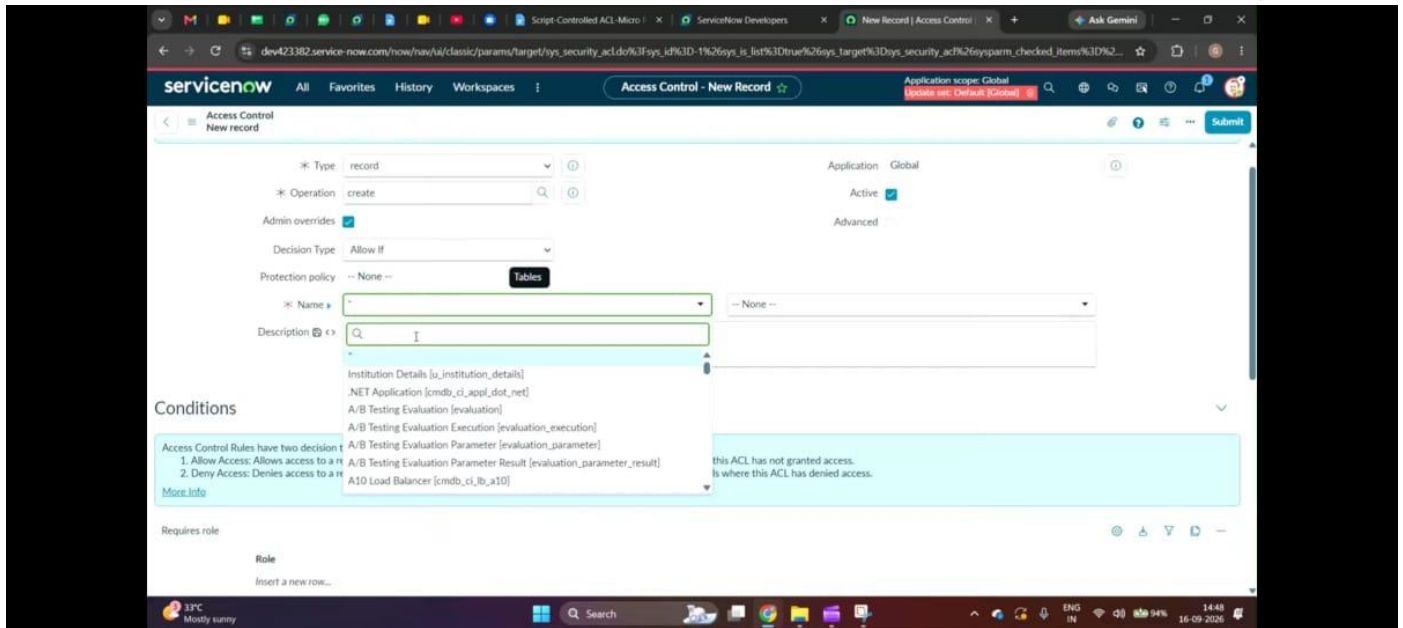
The screenshot shows the ServiceNow 'Institution Details' table. The table has columns: Branch, Description, Email, Faculty Name, Phone Number, Student Name, and Student Roll Number. The 'Branch' column is filtered to show 'CSE', 'ECE', and 'EEE'. The 'ECE' row is selected. The table shows three records:

Branch	Description	Email	Faculty Name	Phone Number	Student Name	Student Roll Number
CSE			Andrew Och		Amos Linnan	STU300
ECE			Abraham Lincoln		Abel Tuter	STU100
EEE			Alejandra Prenatt		Aileen Mottern	STU200

The interface includes a search bar, a 'New' button, and a pagination bar at the bottom showing '1 to 3 of 3'.

Activity 4.2: Verify the Create, Write and Delete ACLs

- A user with the bb2 role can create new Institution Details records; other users cannot.
- A user with the bb3 role can update existing records; other users cannot.
- A user with the bb4 role can delete records; other users cannot.



servicenow All Favorites History Workspaces | Access Control - New Record

Application scope: Global

Update set: Default Subitems

Submit

* Type: record

* Operation: create

Admin overrides: ☒

Decision Type: Allow If

Protection policy: -- None --

* Name: -- None --

Description: -- None --

Conditions

Access Control Rules have two decision types:

1. Allow Access: Allows access to a resource.
2. Deny Access: Denies access to a resource.

[More Info](#)

Requires role

Role: Insert a new row...

Tables

- Institution Details [u_institution_details]
- NET Application [cmdb_ci_app_dot_net]
- A/B Testing Evaluation [evaluation]
- A/B Testing Evaluation Execution [evaluation_execution]
- A/B Testing Evaluation Parameter [evaluation_parameter]
- A/B Testing Evaluation Parameter Result [evaluation_parameter_result]
- A10 Load Balancer [cmdb_ci_lb_a10]

This ACL has not granted access. Is where this ACL has denied access.

33°C Mostly sunny

Search

ENG IN 14:48 16-09-2025

Conclusion:

The Script-Controlled ACL project demonstrates how ServiceNow ACLs can protect a custom table with both role-based and field-based rules. By combining a required role, a data condition (Branch is EEE) and a short JavaScript ACL script, the READ ACL lets administrators see everything, lets bb1 users see only EEE records and denies everyone else, while separate Create, Write and Delete ACLs restrict data changes to the bb2, bb3 and bb4 roles. The result is a simple, maintainable security model that can be reused on other tables and extended to other field values such as ECE or CSE.